

ALEJANDRO DIAZ PAREDES

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SUMMARY

Full-stack software engineer focused on shipping production web applications end-to-end, from database schemas and API architecture to polished, mobile-first interfaces. Comfortable across both traditional server-rendered stacks and modern TypeScript-first frameworks. Hands-on experience integrating LLM workflows, language internationalization, and edge-deployed infrastructure.

RELEVANT SKILLS

- Solid understanding of prompt engineering, and “software as a service” workflow using Agile.
- *(Tools)* Git CLI, GitHub
- *(Languages)* Python, HTML, CSS, JavaScript, Java
- *(Frameworks)* Django, Rest Framework, PostgreSQL, AWS, TypeScript, Next.js

EDUCATION

University of Colorado Colorado Springs, May 2026

Colorado Springs, CO

Bachelor of Science in Computer Science, Software Engineering

PROJECTS

Starview - Full-Stack Astronomy Platform | starview.app | github.com/adiazpar/starview

Django, React, PostgreSQL/PostGIS, Redis Caching, TanStack Query, Mapbox API, AWS, Cloudflare

- Architected and shipped a 48,000-line production codebase with 16 database models, 40+ REST API endpoints, and 66 React components serving a community of stargazers worldwide.
- Achieved 99.3% query reduction (548 to 4 queries) through strategic ORM optimization, cutting page load times by 85%.
- Earned A+ security rating (securityheaders.com) with CSP, HSTS, rate limiting, audit logging, and OAuth 2.0 flows
- PostGis spatial queries for distance-based filtering with great-circle calculations and optimized map clustering for 1,000+ markers.
- Astronomy calculation services integration PyEphem for lunar cycles and raster for processing World Atlas GeoTIFF light pollution data.
- Implemented TanStack Query patterns with optimistic updates, multi-cache invalidation across 8+ related queries, and automatic rollback on errors.
- Integrated 5 external APIs (Mapbox, Open-Meteo, AWS SES, Cloudflare R2, OpenRouteService).
- Implemented i18n for 8 languages and 24-badge gamification system with event-driven automatic rewards.

Kasero - Small Business Management Platform | kasero.app | github.com/adiazpar/kasero

Next.js 15, React 18, TypeScript, Drizzle ORM, SQLite/Turso (libSQL). Tailwind, CSS, framer-motion, OpenAI, fal.ai, Vercel

- Architected a multi-tenant SaaS on edge-deployed libSQL (Turso), spanning 222 TypeScript files and 46 API routes with role-based access (owner/partner/employee) enforced at every write path.
 - Built a fully bilingual (English/Spanish) i18n system with 773 translation keys across 21 namespaces and a typed API message envelope (106 response codes), driven by an automated Claude API translation script.
 - Engineered a snap-to-add AI product pipeline orchestrating a GPT-4o Mini (vision identification), fal.ai Nano Banana (emoji icon generation), and BiRefNet (background removal) in parallel, saving 2-3 seconds per product at ~\$0.04 per captured item.
 - Designed a production barcode system supporting 9 symbologies (EAN-13, UPC-A, Code 128, ITF, Codebar, etc.) with GS1 check-digit validation, canonical 14-digit GTIN deduplication, and live-camera scanning via html5-qrcode with a graceful file-input fallback for desktop.
 - Delivered a three-layer performance strategy - optimistic UI with Lottie animations, a 60-second TTL access cache, and per-business session storage isolation - eliminating redundant API calls across page navigation.
 - Built a mobile-first interface with framer-motion swipe gestures, draggable tab containers, and a PWA manifest enabling install-to-homescreen on iOS and Android.
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WORK EXPERIENCE

Mikron Automation, Controls Engineer Intern, 2024

Englewood, CO

- Developed ladder logic automation following Rockwell standards for manufacturing systems.
- Worked with mechanical and electrical engineering teams to verify installed I/O devices.

University of Colorado Colorado Springs, It Technician I, 2023 - 2025

Colorado Springs, CO

- Cherwell Ticketing system for open incident management and resolution tracking.
 - Resolved 20+ daily service tickets using Azure, Active Directory and Microsoft Intune for 15,000+ university users.
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